Product Requirements Document PRD

Agrinoze

Smart Irrigation System

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Overview

The Agrinoze system is an automated irrigation controller management tool driven by Machine Learning. The system is designed to be seamlessly integrated into various agricultural environments, including greenhouses, open fields, and orchards, supporting all crop types and soil mediums—including conventional, hydroponic, and soilless cultivation.

The system's objective is to ensure ideal conditions within the root zone ecosystem through continuous monitoring and diagnosis of the optimal, on-demand irrigation point.

By integrating real-time data from field sensors - primarily tensiometers and 'Acclima' sensors - the system executes autonomous decision-making. It generates a continuous irrigation plan delivered via precise, measured irrigation pulses throughout the day, eliminating the need for a human-in-the-loop.

Compared to existing evaporation-based protocols, Agrinoze delivers a significant reduction in water consumption (up to 50% savings), renders traditional fertilization obsolete (achieving up to 100% savings), and drives a substantial improvement in both crop yield and quality.

General

Background: The plants grow best with a very high number of very short watering pulses. But there are real world limitations on the number of watering pulses the equipment in the field can produce due to mechanical/hydraulic limitations.

This is a spec for a hardware/software system to control watering and irrigation for plant growing areas.

The system will make and control decisions in real time regarding turning on and off the irrigation (open/close the watering solenoid).

Systems Outputs:

- Hardware: Turn faucet solenoid/tap on off.
- Screen/UI:
  - Graphs showing sensor readings.
  - Alerts for the user/administrator.
  - Reports (e.g. water consumption, historical sensor readings, etc').

System Inputs:

- Various sensors in the ground (described later)
- system clock.
- Ground type (presets).
- Watering capabilities of the drippers (maximum achievable discharge e.g. 1L/hour drippers)
- Available water type (regular/ distilled)
- Actual water discharge achieved vs planned (clogs/leaks),

The system is designed to create optimal growing conditions and reduce water consumption.

The main component of the system, and the most important measurement, is the VWC parameter. Each type of soil has an Optimal VWC % that can be reached. This optimal VWC provides the best growing conditions and reaching it is the ultimate goal for the system.

Terminology

1) **Cycle** - Duration of time in which a repeating/cyclical irrigation program runs. Currently the cycle is one day / 24 hours
2) **Pulse** - A single instance of irrigation., from the time when the water is turned on until the time it is turned off.
3) **Pulse Duration** - Duration of time for which the water is ON. For example, 1 minute.
4) **Interval** - Amount of time between neighboring pulses of irrigation.
   An interval is composed of X minutes water on + Y minutes water off.
   The interval is defined as
   Cycle duration / number of pulses.
   (one iteration and duration of time in which the water is OFF+ON (e.g. 7.2 minutes)
5) **Number of pulses** - number of pulses performed in a cycle. For example, 1 cycle could be 24 hours. Number pulses = 200 pulses per day. Interval, Cycle and pulses are all related
6) **Irrigation Program** - Combination of Cycle, Pulses and Intervals.
   Interval duration * Number of pulses = cycle.
7) **VWC - Volumetric Water Capacity**. Measures humidity in the ground, scale 0-100%.
   Measured as total water content in the soil divided by the total volume of the soil. Mathematically:

$$\boldsymbol{VWC = \frac{V_w}{V_s}}$$

   $\boldsymbol{V_w}$ (Volume of Water)
   V_s (Total Soil sample Volume / Bulk Volume)
8) **Stages** - Different soil phases which create different programs for the system which affect the way it calculates the irrigation. The soil is supposed to transition from one stage to the next because of the way the irrigation program affects it.
   3 main stages for reference - Initial stage (before starting irrigation), Calibration Stage, Optimization stage.
9) **Soil type**: The kind of soil found at the irrigation site (Medium/heavy etc.). Soil type is given per site, where the system is intended to be implemented. Soil type presets:
   - **Heavy Soil** - 50% and more.
   - **Medium Soil** - 40-50%.
   - **Sandy Soil** - 30-35%.
   - **Soilless** (artificial bedding) - 30-50%.

   Note - The given percentages are Optimal VWC rage for each soil type.

10) **Tensiometer** - System Sensor to measure surface tension of water in soil. There are two different tensiometers in the system - one located at depth of 20 cm and one at 40cm. Measurement is in millibar.
11) **Tensiometer Depth** - how deep a tensiometer sensor is located in the soil. There are two sensors: One at 20cm depth, and one at 40cm depth.
12) **Drainage** - capability of soil to remove excess water. Lack of drainage can destroy the plants' growth and needs to be managed.
13) **Saturation** - Condition when soil is completely filled with water. This is a bad condition which hurts plants. Tensiometer measures 0 millibar in this situation.
    (move this) This condition should not be possible in Soilless ground.
    The danger is when adding water to reach optimal VWC of hitting the undesired saturation in the process. For this reason, we will sometimes change/modify the watering plan to differentiate from the one designed for ultimate VWC. The system tries to juggle both VWC and tensiometer readings to achieve an optimal watering system.
14) **Irrigation Controller** - A piece of hardware that allows the system to:
    - Control (start/stop) the irrigation.
    - Read sensor readings/parameters from the soil.
    - Read health of the sensors (faulty/working/unplausible readings).
    - Monitor suspiciously high-water discharge which can indicate leaks/clogged drippers/taps or pipelines.
    - **Note** - In the first software development stage of our project, this controller is simulated by software. The system will take inputs and provide outputs to a controller mockup component visually or from file. Once we provide a physical controller (on a later project phase) - we will need to connect the Controller with the algorithm via an API layer.
15) **Discharge** - volumetric rate of flow/flow rate. The rate of water discharge that the mechanical/hydraulic irrigation components can provide/deliver to the soil. (e.g. 30 cubic per hour). We will monitor **Planned Flow** and **Actual Flow** during system operation. These calculations require the specifications from the hydraulic system designer.
16) **Control Screen** - allows the administrator (e.g. Agronomist) to control various functions of the system, manual override, setting the inputs such as discharge, soil type, water type, pulse duration etc.
17) **Monitoring Screen -** allows an on-site system user (e.g. Farmer) to view system and soil state, alerts, sensor readings, without the ability to control the system. The information will be text only regarding irrigation outputs throughout the day/cycle, and general faults (discharge mismatch only).

System View

1. **Sensors**:
   - **Tensiometer** X 2, called 20cm 40cm. produces values in mb.
     Values vary from 0 mb = Saturation, up to 200 mb
     Values above that are suspect as a malfunction sensor.
   - **'Acclima' Sensor** – a high-precision, professional-grade **soil moisture and environmental sensor** primarily used in agriculture, research, commercial turf management, and advanced greenhouse cultivation which Measures (mainly):
     - **VWC**.
     - **EC Bulk** - The actual electrical conductivity of the bulk soil or substrate (the soil-water-air mixture).
     - **EC Pore** - The electrical conductivity of the water present in the soil pores.
     - **Soil Temperature**.
   - **pH meter** - an electronic instrument used to measure the hydrogen-ion activity in a water-based solution, indicating its acidity or alkalinity

   Note - Only VWC and Tensiometer (depth 20cm) will control the irrigation. Other sensor readings/parameters will only be displayed on the administrator control screen for viewing.
2. **Irrigation controller**.
3. **Hydraulic system** on site – taps/solenoids, watering pipelines, drippers. The hydraulic system specs (water discharge/flow rate etc.) are given at the beginning of the implementation.
4. **Control Screen**.
5. **Monitoring screen**.
6. **Note** - Software Architecture, Tech stack, cloud based, design principles - TBD later.

Functional Requirements

Inputs:

1. Remote System Control - not currently relevant
2. Receiving system inputs from sensors - Sensors are read every 10 minutes by the program.
3. Sensor Data History - The system is required to record and store the history of sensor data. Sensor history backlog is kept for 3 years.
4. Cycle duration can be changed. Currently it is set 24 hours.
5. Ability to set the pulse duration for various stages of the irrigation program.
6. Ability to set the interval between pulses.
7. Variables that need to be configurable by the administrator. In-order to do this we will need to have names for them. E.g.
    - Calibration Sub-Stage 1:
      - Number-of-pulses: 200
      - Pulse duration: 2 minutes
    - Calibration Sub-Stage 2:
      - Number-of-pulses: 180
      - Pulse duration: 30 seconds

Output:

1. Irrigation program - The irrigation program will run continuously and modify various parameters such as number of pulses and pulse duration – derived by the readings of the sensor data and the algorithm (described later in the Irrigation algorithm chapter).
   This continuous program will control the irrigation commands to the watering/hydraulic system - through the controller.
2. Display graphs of sensor readings on control/administrator screen. Shows sensor data values at sampled points over the past 24 hours timeline.
3. Water Usage: The system must be able to report how much water has been used per day, by comparing the planned discharge to the actual discharge reading from the controller, and alert if there is a mismatch of too much discharge above/below 1% of planned. This will be checked every 2 hours. This may be calculated by either:
   A: By calculating: (Discharge rate/hour per dripper) (*number of pulses per day) * (pulse duration) (* days)(* number of drippers).
   This is reported per land/field area (e.g. per hectare)

B: The system can use a controller's function to get the water readings (where applicable).

4. Alerts module - the system will provide 2 types of alerts to the user & administrator, by the severity of the situation.
    a. **Yellow** alert – pay attention, no immediate actions is needed.
    b. **Red** alert – An action should be taken by the admin and/or user, in-order to fix the situation.

Irrigation Algorithm

1. The irrigation program runs in several main stages, which may contain sub-stages.
    - **Stage 1** - **Initial stage** - before starting irrigation.
    - **Stage 2** - **Calibration stage** – Drenching.
    - **Stage 3** - **Optimization/Continuous Irrigation**.
2. **Initial stage**:
    - The system continuously reads the 20 cm depth tensiometer mb value.

      Note - If at any time, regardless of stage, the 40cm tensiometer falls below 10mb alert the administrator. Even during the optimization stage.

3. **Calibration Stage** - composed of **3 sub-stages**:

   3.1. **Calibration Sub-Stage 1**:
    - The system will perform an irrigation program of 200 pulses of 2-minute irrigation.
    - At 200 pulses and 1440 minutes per 24hours, each pulse + interval = 7.2 minutes. With a pulse of 2 minutes, this means 2 min water on, 5.2 minutes water off.
    - After 120 hours, 5 days, we expect to see decreasing tensiometer readings (water tension) For example 50,48,45,40.
    - If Tensiometer reading does not drop below 40mb after 120 hours (5 days), give an administrator alert.
    - The system will continue at 200 pulses of 2 minutes, until the tensiometer drops below 10 millibars for 3 consecutive days.
    - In the range between 40-10 we are still at stage 1.
    - Calibration Sub-stage 2 will not begin until tensiometer reading drops below 10mb for 3 consecutive days.
    - Once the readings drop below 10mb for 3 consecutive days (trigger) we change the program.

   3.2. **Calibration Sub-Stage 2**:
    - The system changes the program to 180 pulses per 24hours, intervals of 8 minutes with 30 second pulses. Which means 30 seconds of irrigation and 7.5 minutes pause.
    - In some areas the Administrator can change this to 180 pulses, with 1 minute irrigation and 7 minutes pause for sites with significant height difference between drippers. This will be an

administrator-controlled variable. The administrator will determine this in the initial stages of site set up.

- This program will continue for up to 2 weeks - but the next stage may come sooner.
- The next stage will come when the system recognizes tensiometer rise above 40 mb for at least 3 consecutive days. If this does not happen within two weeks (14 days): Alert Administrator.
- If reading drops below 10 for 3 days again - Alert Administrator.

3.3. **Calibration Sub-Stage 3:**

- This is a repetitive stage until we find stable behavior.
- On each iteration the system samples the readings at the end of two weeks.
- We do not make any changes during these two weeks except for manual intervention (that the Administrator can make), after alerts.
- During these two weeks the system will give yellow alerts if:
    i. The tensiometer reading falls below 10mb for 3 days.
    ii. The tensiometer reading rises above 40 mb for 3 days.
- At the end of each two weeks' iteration:
    i. If reading drops below 10mb we reduce the number of pulses by 20 without changing pulse duration. (e.g. change 180 to 160 pulses 9-minute interval) with 30 seconds irrigation.
       If reading rises above 40mb we increase the number of pulses by 20, without changing pulse duration.
    ii. If reading rises above 40mb we keep the same number of pulses and increase the irrigation time by 30 seconds.
- We exit this above 'loop'/iteration (and exit the stage) when we recognize a stable reading of values between 20mb and 40mb for 2 consecutive weeks.
- During the waiting time for 20mb to 40mb range we need to monitor VWC. If VWC does not rise by 10% after 1 week, give an administrator alert. For example, if the initial VWC was 27% and it did not rise to at least 37% give an alert.
- After two weeks, when the reading is between 20mb and 40mb we exit the calibration stage and enter continuous optimization stage.

4. **<u>Continuous Optimization Stage</u>:**

- The system pulse length always remains 30 seconds at the optimization stage.
- At the end of every 2 week cycle the system will check VWC.
  The optimal VWC is per ground type
  - Heavy Soil - 50% and more
  - Medium Soil - 40-50%
  - Sandy Soil - 30-35%
  - Soilless (artificial bedding) - 30-50%
- If VWC > optimal (the minimal value# of each soil optimal range) for the ground -don't do anything.
  If VWC < the minimal value# of the optimal range - add 20 pulses for the next two weeks.
- In parallel to the above - we also continue the 10mb-40mb tensiometer checks and increase by 20 pulses if it climbs over 40mb and decrease by 20 pulses if falls under 10mb.
  Note - There is no double increase: if both tensiometer rises above 40 and VWC fall below optimal, do not add 40 pulses, add only 20 pulses.
  If the tensiometer falls under 10 mb we add 20 pulses even if VWC is > optimal.
- If at any time, regardless of stage, the 40cm tensiometer falls below 10mb alert the administrator. Even during optimization stage.
- If at any time the 20cm tensiometer falls below 1 mb, give a Yellow alert.
- If at any time, regardless of stage, the 20cm tensiometer falls below 1 mb for 3 days – give a Red alert (Warning) to the administrator.